TIME & WORK

Questions followed by worked solutions

Each question is followed immediately by its step-by-step solution. Exam names, question-number headings, and separate final-answer lines have been removed.

Where the original wording is incomplete or ambiguous, the solution flags that limitation rather than inventing missing information.

A takes 10 days and B takes 25 days to complete a job. How long do they take together?

Worked solution
Take the whole job as 1. A's rate = 1/10 job/day; B's rate = 1/25. Together: 1/10 + 1/25 = 5/50 + 2/50 = 7/50 job/day. Time = 1 ÷ (7/50) = 50/7 days.

Hunny can finish a job in 10 days. After working alone for 4 days, Abhinav joins him and they finish the remaining work in 2 more days. Find Abhinav's solo time.

Worked solution
Hunny's rate = 1/10. In 4 days he completes 4/10 = 2/5, leaving 3/5. Their combined rate over the next 2 days is (3/5)/2 = 3/10 per day. Abhinav's rate = 3/10 – 1/10 = 1/5. So Abhinav alone takes 5 days.

Aashna, Risitha and Vineet together can do a job in 15 days. They work together for 2 days; then Aashna leaves. Risitha and Vineet work 10 more days; then Risitha leaves. Vineet finishes the job in another 40 days. Vineet alone takes 75 days. Find Aashna's solo time.

Worked solution
Let rates be A, R and V. A+R+V = 1/15 and V = 1/75. The whole job is 2(A+R+V) + 10(R+V) + 40V = 1. Thus 2/15 + 10(R+V) + 40/75 = 1, so 10(R+V) = 1 – 2/15 – 8/15 = 1/3; hence R+V = 1/30. Therefore R = 1/30 – 1/75 = 1/50. A = 1/15 – 1/30 = 1/30.

A and B alone finish a job in 40 and 120 days. They work together for 10 days, then extra work equal to what B could do in 40 days is added. C joins A and B, and the remaining original plus additional work is completed in 5 days. Find C's time for the original job.

Worked solution
A+B rate = 1/40 + 1/120 = 1/30. In 10 days they do 1/3 of the original job, leaving 2/3. Added work = 40 × (1/120) = 1/3 of the original job. Total work remaining after the addition = 2/3 + 1/3 = 1 whole-job equivalent. In the final 5 days A+B do 5/30 = 1/6, so C does 1 – 1/6 = 5/6 in 5 days. C's rate = (5/6)/5 = 1/6 job/day; time = 6 days.

A can do a job in 20 days and B in 30 days. They work together for 7 days, then C alone finishes the remainder in 10 days. How long would C take alone for the whole job?

Worked solution
A+B rate = 1/20 + 1/30 = 1/12. In 7 days they complete 7/12, leaving 5/12. C completes 5/12 in 10 days, so C's rate = 1/24 job/day. Therefore C alone takes 24 days.

A, B and C alone take 16, 32 and 48 days. They start together; B leaves 8 days before completion and C leaves 6 days before completion. Find the total duration.

Worked solution
Let the job take t days. A works t days, B works t–8 days and C works t–6 days. Therefore t/16 + (t–8)/32 + (t–6)/48 = 1. Multiply by 96: 6t + 3t – 24 + 2t – 12 = 96, so 11t = 132 and t = 12.

Ram can build a wall in 30 days and Laxman in 60 days. When together, each person's efficiency drops to 80% of usual. Find their joint time.

Worked solution
Usual combined rate = 1/30 + 1/60 = 1/20. At 80% efficiency, the combined rate is 0.8 × 1/20 = 1/25. Time = 25 days.

Gee-k takes 20 days and Rishi 30 days. With Prateek, the job is finished in 8 days. The total payment is ■5,550. Find Prateek's share if payment is proportional to work.

Worked solution
Prateek's rate = 1/8 – 1/20 – 1/30 = 1/8 – 1/12 = 1/24. In 8 days, his share of the work is 8/24 = 1/3. His payment = ■5,550 × 1/3 = ■1,850.

Anil alone takes 20 days and Sunil 40 days. Anil starts; Sunil joins after 3 days. Bimal later joins, and Bimal completes 10% of the job. If the job takes d days in total, find d.

Worked solution
Anil works d days, Sunil works d–3 days, and Bimal contributes 1/10 of the job. So d/20 + (d–3)/40 + 1/10 = 1. Multiply by 40: 2d + d – 3 + 4 = 40, giving 3d = 39 and d = 13 days.

Amal takes 10 days, Bimal 8 days, and all three including Kamal finish in 4 days. They share ■1,000 in proportion to work. Find Kamal's share.

Worked solution
Combined rate = 1/4. Amal's rate = 1/10 and Bimal's = 1/8. Kamal's rate = 1/4 – 1/10 – 1/8 = 10/40 – 4/40 – 5/40 = 1/40. In 4 days Kamal does 4/40 = 1/10 of the job, so his share is ■1,000 × 1/10 = ■100.

Two painters together paint a wall in x minutes. B alone takes 5 minutes longer than x. A alone takes 40 minutes longer than B. Find x.

Worked solution
B's time = x+5 and A's time = x+45. Their combined rate is 1/x, so 1/x = 1/(x+5) + 1/(x+45). Combining fractions gives (x+5)(x+45) = x(2x+50). Expanding: $x^2+50x+225 = 2x^2+50x$, so $x^2 = 225$ and the positive solution is x = 15.

Two men and four boys finish a job in 8 days; four men and two boys finish it in 6 days. How long would one man and one boy take together?

Worked solution
Let one man's daily rate be m and one boy's be b. Then 2m+4b = 1/8 and 4m+2b = 1/6. Simplify: m+2b = 1/16 and 2m+b = 1/12. Solving gives b = 1/72 and m = 5/144. Thus m+b = 7/144, so the time is 144/7 days.

A job is completed by 11 men and 16 boys in 2 days, or by 5 men and 11 boys in 4 days. Find the time for 1 man and 4 boys.

Worked solution
Let the total work be W, with daily rates m and b. 11m+16b = W/2 and 5m+11b = W/4. Hence 11m+16b = 2(5m+11b), giving m = 6b. Then W = 4(5m+11b) = 4(30b+11b) = 164b. The rate of 1 man and 4 boys is m+4b = 10b. Time = 164b/(10b) = 16.4 days.

15 humans and 5 robots finish a job in 30 days; 5 humans and 15 robots finish it in 60 days. Find the time for 15 humans alone.

Worked solution
Let daily rates be h per human and r per robot. Then 15h+5r = 1/30 and 5h+15r = 1/60. Multiply the first equation by 3: 45h+15r = 1/10. Subtract the second equation: 40h = 1/10–1/60 = 1/12, so h = 1/480. Fifteen humans work at rate 15/480 = 1/32, taking 32 days.

Ten men and 12 children finish a job in 10 days. Each child takes three times as long as a man, so a child's rate is one-third of a man's. How long would 12 men take?

Worked solution
Let a man's daily rate be m; a child's rate is m/3. The daily team rate is 10m + 12(m/3) = 14m. Total work = 10 × 14m = 140m. Twelve men work at 12m per day, so time = 140m/(12m) = 35/3 days.

Five men and eight women finish a job in 8 days; four men and six women finish it in 10 days. How long would ten women take?

Worked solution
Let daily rates be m and w. Then 5m+8w = 1/8 and 4m+6w = 1/10. Multiplying the first by 2 gives 10m+16w = 1/4; multiplying the second by 2.5 gives 10m+15w = 1/4. Subtraction gives w = 0. This implies women contribute no work under the stated equations, so ten women alone cannot finish the job. The supplied option 'None of these' is consistent with the data, although the question's numbers are unusual.

Anil alone paints a house in 60 days and Bimal in 84 days. Anil works 10 days alone; then all three finish in 14 more days. The total payment is ■21,000. Find Charu's proportional share.

Worked solution
Anil completes 10/60 = 1/6 before the others join, leaving 5/6. Their combined rate for the final 14 days is (5/6)/14 = 5/84. Charu's rate = 5/84 − 1/60 − 1/84 = 13/420. In 14 days Charu does 14 × 13/420 = 13/30 of the job. His share = ■21,000 × 13/30 = ■9,100.

Amar and Akbar finish in 12 months, Akbar and Anthony in 16 months, and Anthony and Amar in 24 months. Find the solo time of the person who is neither fastest nor slowest.

Worked solution
Monthly pair rates are 1/12, 1/16 and 1/24. Amar's rate = [(A+B)+(A+C)−(B+C)]/2 = (1/12+1/24−1/16)/2 = (4/48+2/48−3/48)/2 = 1/32 per month. Akbar's rate = (1/12+1/16−1/24)/2 = 5/96; Anthony's = (1/16+1/24−1/12)/2 = 1/96. Akbar is fastest, Anthony slowest, so Amar is the middle worker and takes 32 months.

Anil paints a house in 12 days and Barun in 16 days. They and Chandu finish it together in 6 days for ■24,000. Find Chandu's share proportional to work.

Worked solution
Combined rate = 1/6. Anil and Barun's rates sum to 1/12+1/16 = 7/48. Chandu's rate = 1/6−7/48 = 1/48. In 6 days Chandu completes 6/48 = 1/8 of the job. His payment is ■24,000 × 1/8 = ■3,000.

Ram alone erects a wall in 16 days and Samar in 12 days. They work on alternate days, Ram first. Find the completion time.

Worked solution
In each 2-day cycle they complete 1/16+1/12 = 7/48. In 12 days (6 cycles), they complete 42/48 = 7/8. On day 13 Ram completes 1/16 = 3/48, bringing the total to 45/48 = 15/16. The remaining 1/16 is completed by Samar at rate 1/12, requiring (1/16)/(1/12) = 3/4 day. Total = 133/4 days = 55/4 days.

A and B together build 221 m in 100/9 days. Alternating days, A starts and they build the same wall in 89/4 days. For a wall twice as long they earn ■1,800. Find B's share.

Worked solution
Together, their rate is 221 ÷ (100/9) = 1989/100 m/day. In 89/4 = 22¼ days, A works 11¼ days and B works 11 days. Thus 11¼A + 11B = 221, while A+B = 1989/100. Substituting A = 1989/100−B gives 11.25(1989/100−B)+11B = 221, which yields B/(A+B) = 5/9. For a wall twice as long, total pay is ■1,800; B receives 5/9 × ■1,800 = ■1,000.

Anu, Vinu and Manu alone take 15, 12 and 20 days. Vinu works daily; Anu works on odd days and Manu on even days. Find the completion time.

Worked solution
On an odd day, Anu+Vinu complete 1/15+1/12 = 3/20. On an even day, Vinu+Manu complete 1/12+1/20 = 2/15. Each 2-day block completes 3/20+2/15 = 17/60. After 6 days, three blocks complete 51/60 = 17/20. Day 7 is an odd day and adds 3/20, finishing the job.

Six men working 5 days for 8 hours per day complete a task. How many men are needed to finish half that task in 4 days at 6 hours per day?

Worked solution
Full task requires 6×5×8 = 240 man-hours. Half requires 120 man-hours. Each new worker supplies 4×6 = 24 hours, so required workers = 120/24 = 5.

Twelve examiners work 16 hours per day for 18 days to check 24,000 sheets. How many hours per day should 24 examiners work for 36 days to check 36,000 sheets?

Worked solution
Work is proportional to examiners × hours/day × days. Let h be the required hours. (24×h×36)/(12×16×18) = 36,000/24,000 = 3/2. Since 24×36/(12×18)=4, we get 4h/16 = 3/2, so h = 6 hours/day.

A contractor employs 75 men for 8 hours/day for 150 days. After 90 days only 2/7 of the work is complete. How many additional men are needed for the remaining 60 days at 10 hours/day?

Worked solution
Remaining work is 5/7, while completed work is 2/7, so remaining/completed = 5/2. The remaining work requires 75×8×90×(5/2) = 135,000 man-hours. In 60 days at 10 hours/day, each man supplies 600 hours, so total men required = 135,000/600 = 225. Additional men = 225–75 = 150.

Fifteen men or 24 women or 36 boys complete a job in 12 days at 8 hours/day. How many men must work with 12 women and 6 boys to finish a job 9/4 as large in 30 days at 6 hours/day?

Worked solution
Equivalent rates: 15 men = 24 women = 36 boys, so 12 women = 7.5 men and 6 boys = 2.5 men. The new job is 9/4 as large, but the available hours per worker are 30×6 instead of 12×8. Required workforce in men-equivalents = 15×(9/4)×(12×8)/(30×6) = 18. Existing 12 women + 6 boys equal 7.5+2.5 = 10 men. Additional men required = 18–10 = 8.

A contractor must build a 6 km road in 200 days using 140 people. After 60 days, only 1.5 km is complete. How many additional people are needed to finish on time?

Worked solution
After 60 days, 1.5 km is built, so the remaining 4.5 km is three times as much work. The remaining time is 140 days, which is 140/60 times the elapsed time. Required workforce = 140 × (4.5/1.5) × (60/140) = 420 people. Additional people = 420–140 = 280. This conflicts with the supplied answer key of 40; the calculation from the stated figures gives 280.

A group completes 35% of a project by working 7 hours/day for 10 days. Then 10 people leave; the rest complete the remaining 65% in 14 days at 10 hours/day. Find the original group size.

Worked solution
Let the original group size be n and each person's hourly rate be constant. Work is proportional to people × hours × days. Thus 35% is proportional to n×7×10, and 65% to (n–10)×10×14. Divide the second by the first: (65/35) = [(n–10)×140]/[n×70] = 2(n–10)/n. So 13/7 = 2(n–10)/n; 13n = 14n–140, giving n = 140. The source question's final phrase is truncated, so this solves for the implied original group size.

A person can finish a job in 120 days. One person works on day 1; a new equally efficient person joins each successive day. How many days are needed?

Worked solution
On day 1, 1 person works; day 2, 2 people; and so on. In n days, total work is (1+2+…+n)/120 = n(n+1)/(240). Set n(n+1)/240 = 1, giving n(n+1)=240. Since 15×16=240, the job finishes on day 15.

A and B together finish a task in 12 days. If A works at half her usual rate and B at triple his usual rate, they finish in 9 days. Find A's solo time.

Worked solution
Let usual rates be a and b. a+b=1/12 and a/2+3b=1/9. Substitute a=1/12−b: 1/24−b/2+3b=1/9, so (5/2)b=1/9−1/24=5/72, giving b=1/36. Therefore a=1/12−1/36=1/18. A alone takes 18 days.

Rahul starts at 9 AM; Gautam joins at 11 AM; they finish at 5 PM. If both had started at 9 AM, they would finish at 4:30 PM. Find Rahul's solo time.

Worked solution
Actual work: Rahul works 8 hours and Gautam 6 hours, so 8r+6g=1. If both work together for 7.5 hours, (r+g)=1/7.5=2/15. Substitute g=2/15−r into the first equation: 8r+6(2/15−r)=1, so 2r=1−4/5=1/5 and r=1/10. Rahul alone takes 10 hours.

Ramesh and Ganesh together finish in 16 days. After 7 days, Ramesh's efficiency falls by 30%; the job finishes in 17 days total. How long would Ganesh alone take to finish the work remaining after day 7?

Worked solution
Let original rates be r and g. r+g=1/16. After day 7, 10 days remain and the unfinished work is 1−7/16=9/16. Thus g+0.7r=(9/16)/10=9/160. Subtract this from r+g=10/160: 0.3r=1/160, so r=1/48 and g=1/16−1/48=1/24. Ganesh alone needs (9/16)/(1/24)=13.5 days for the remaining work.

Three men and eight machines finish a job in half the time taken by eight men and three machines. Two machines finish the job in 13 days. How many men finish it in 13 days?

Worked solution
Let a man's rate be m and a machine's rate be x. Since the first group takes half the time, its rate is twice the second group's rate: 3m+8x=2(8m+3x). Hence 2x=13m, so m=2x/13. Two machines complete the job in 13 days, so total work = 26x (machine-rate days). If n men complete it in 13 days, work = 13nm = 13n(2x/13)=2nx. Set 2nx=26x to get n=13.

B takes 25% longer than A to do a job. A does half the job alone, then both work together for 4 days, and B finishes the final 5%. The total elapsed time is 13 days. Find B's solo time.

Worked solution
Let A's solo time be t; B's is 1.25t. Their rates are 1/t and 4/(5t). A completes half the job alone, the pair work for 4 days, and B completes the final 5%. Thus 1/2 + 4(1/t+4/(5t)) + 1/20 = 1. Therefore 4×9/(5t)=9/20, so 36/(5t)=9/20 and t=16 days. B takes 1.25×16=20 days. The elapsed time is 8 days for A to do half + 4 joint days + 1 day for B to do 5% = 13 days, consistent with the question.

Neeta and Geeta together earn in one day what Sita earns in 6 days. Sita and Neeta together earn in one day what Geeta earns in 2 days. Find the ratio of the highest daily earning to the lowest.

Worked solution
Let daily earnings be N, G and S. N+G=6S and S+N=2G. From the second, N=2G−S. Substitute into the first: 3G−S=6S, so G=7S/3. Then N=2(7S/3)−S=11S/3. Neeta earns the most and Sita the least, so the ratio is 11:3.

Solo completion times of Anu, Tanu and Manu are in the ratio 5:8:10. Together they finish in 4 days at 8 hours/day. Anu and Tanu instead work together for 6 days at 6 hours 40 minutes/day. How many more hours does Manu need to finish the remainder?

Worked solution
Let their hourly rates be c/5, c/8 and c/10. In 32 hours together, work is 32c(1/5+1/8+1/10)=1. Since the bracket is 17/40, c=5/68. Anu and Tanu work 6×(62/3)=40 hours, doing 40c(1/5+1/8)=13c. Total work is 32×17c/40=13.6c, so the remainder is 0.6c. Manu's rate is c/10, so remaining time = 0.6c ÷ (c/10)=6 hours.

Bob alone finishes in 40 days. Alex is twice as fast as Bob and three times as fast as Cole. The pairs work in this repeating order: Alex+Bob, Bob+Cole, Cole+Alex. How many days does Alex work before completion?

Worked solution
Rates: Bob=1/40, Alex=1/20, Cole=1/60. The three-day roster completes (1/20+1/40)+(1/40+1/60)+(1/60+1/20)=3/40+1/24+1/15=11/60. After 15 days (five cycles), 55/60=11/12 is done. On day 16, Alex+Bob complete 3/40=9/120, leaving 1/120. Alex has worked on days 1, 3, 4, 6, … and day 16; by the end of day 16, Alex has worked 11 days. The remaining 1/120 is finished early on day 17 by Bob+Cole, when Alex is not working.

Gautam and Suhani together finish a job in 20 days. When Gautam does only 60% of his usual work, Suhani must do 150% of hers to compensate exactly. Find the solo time of the faster worker.

Worked solution
The reduction in Gautam's output is 40% of his usual rate; Suhani's extra output is 50% of hers. Equal compensation means 0.4G=0.5S, so G:S=5:4. Their combined rate is 1/20, or 9 equal parts, so Gautam's rate is (5/9)(1/20)=1/36. Gautam is faster and takes 36 days alone.

John takes twice as long as Jack. Jack and Jim together finish in one-third of John's solo time. John takes 3 days longer than all three working together. Find Jim's solo time.

Worked solution
Let Jack's rate be j. John's rate is j/2. Since Jack+Jim finish in one-third of John's time, their combined rate is 3(j/2)=3j/2, so Jim's rate is 3j/2–j=j/2. All three together work at j/2+j+j/2=2j. Their time is 1/(2j), while John's time is 2/j. The difference is 3 days: 2/j–1/(2j)=3, so 3/(2j)=3 and j=1/2. Jim's rate is 1/4, so Jim takes 4 days.

Twelve men can complete a job in 8 days. After 3 days, 3 more men join. How long do the 15 men need to finish the remaining work?

Worked solution
Total work = 12×8=96 man-days. In the first 3 days, 12 men do 36 man-days, leaving 60. With 15 men, remaining time = 60/15=4 days. (Total elapsed time would be 3+4=7 days.)

Pipes A and B fill a tank in 36 and 45 minutes; C empties it in 30 minutes. All are opened, and the emptying pipe is closed after 6 minutes. Find total filling time.

Worked solution
Net rate while all are open = 1/36+1/45–1/30 = (5+4–6)/180 = 1/60 tank/min. In 6 minutes they fill 1/10. Remaining = 9/10. After C closes, A+B rate = 1/36+1/45=1/20, so remaining time = (9/10)/(1/20)=18 minutes. Total elapsed time = 6+18=24 minutes.

Pipes A and B fill a tank in 5 and 6 hours; C empties it in 12 hours. Find the time with all three open.

Worked solution
Net rate = 1/5+1/6–1/12 = (12+10–5)/60 = 17/60 tank/hour. Time = 60/17 hours, about 3.53 hours.

Three pipes fill a reservoir in 10, 15 and 20 hours. They are opened one after another at equal time intervals; the reservoir fills in 5 hours. Find the interval.

Worked solution
Let the interval be x hours. The first pipe works 5 hours, the second 5–x, and the third 5–2x. Total work: 5/10+(5–x)/15+(5–2x)/20=1. The constant terms sum to 13/12, and the x terms sum to –x/6. So 13/12–x/6=1, giving x/6=1/12 and x=1/2 hour.

Four inlet pipes each fill a tank in 4 hours; two outlet pipes each empty it in 3 hours. Find the time if all six are open.

Worked solution
Four inlets' rate = 4×1/4=1 tank/hour. Two outlets' rate = 2×1/3=2/3 tank/hour. Net rate = 1–2/3=1/3 tank/hour, so the tank fills in 3 hours.

An inlet fills a tank in 30 hours. Five identical inlets and four identical outlets together fill it in 10 hours. How long does one outlet take to empty half a tank?

Worked solution
Inlet rate = 1/30. Let an outlet rate be o. Then 5/30–4o=1/10. Thus 1/6–1/10=4o, so 1/15=4o and o=1/60 tank/hour. To empty half a tank takes (1/2)/(1/60)=30 hours.

An inlet fills a tank in 8 hours. With the outlet open, the inlet fills it in 10 hours. How long does the outlet alone take to drain a full tank to half-full?

Worked solution
Inlet rate = 1/8 and net rate with outlet open = 1/10. Outlet rate = 1/8–1/10=1/40 tank/hour. Draining half a tank takes (1/2)/(1/40)=20 hours.

A tank fills in 6 hours with 6 identical filling pipes and 5 draining pipes, and in 60 hours with 5 filling pipes and 6 draining pipes. Find the time with 2 filling pipes and 1 draining pipe.

Worked solution
Let a filling pipe's rate be f and a draining pipe's rate be d. 6f–5d=1/6; 5f–6d=1/60. Subtracting gives f+d=3/20. Solving with the first equation gives f=1/12 and d=1/15. Then 2f–d=1/6–1/15=1/10 tank/hour, so the tank fills in 10 hours.

Type A and B inlets have constant rates. Ten A and 45 B fill a tank in 30 minutes; 8 A and 18 B fill it in 60 minutes. Find the time for 7 A and 27 B.

Worked solution
Let the rates be a and b tanks/minute. 10a+45b=1/30 and 8a+18b=1/60. Solving gives a=1/1200 and b=1/1800. Thus 7a+27b=7/1200+27/1800=1/48 tank/minute, so filling time is 48 minutes.

A and B together finish a job in T days. A alone takes T+5 days and B alone T+45 days. Find T.

Worked solution
1/T=1/(T+5)+1/(T+45). Multiply by T(T+5)(T+45): (T+5)(T+45)=T(2T+50). Expand: $T^2+50T+225=2T^2+50T$, so $T^2=225$. Since time is positive, T=15.

Amit is 331/3% more efficient than Sumit. Amit works x days and Sumit completes the remaining work in x+3 days. Their work amounts are in the ratio 2:3. Find x.

Worked solution
Efficiency ratio Amit:Sumit = 4:3. Work ratio is (4x):(3(x+3))=2:3. Cross-multiply: 12x=6(x+3), so 12x=6x+18 and x=3.

A and B together finish a job in 7.5 days. If A completes half alone and then B completes half alone, total time is 20 days. A is faster. Find B's solo time.

Worked solution
Let solo times be a and b. Half-job times sum to 20, so a/2+b/2=20, hence a+b=40. Their combined rate is 1/a+1/b=1/7.5=2/15. Since (a+b)/(ab)=2/15, 40/(ab)=2/15, giving ab=300. The numbers with sum 40 and product 300 are 10 and 30. A is faster, so a=10 and b=30.

Abdul, Bimal, Charlie and Dilbar take 10, 12, 15 and 18 days alone. Which listed attendance pattern completes exactly half the job in 3 days?

Worked solution
Check option E: Abdul works 1 day, Bimal 2 days, Charlie 1 day, Dilbar 3 days. Work = 1/10 + 2/12 + 1/15 + 3/18. With denominator 180 this is 18/180+30/180+12/180+30/180=90/180=1/2. Therefore option E is the possible pattern.

X, Y and Z alone finish a job in 8, 12 and 30 days. At most two people can work per day, and nobody may work more than two consecutive days. Find the minimum completion time.

Worked solution
Rates are X=1/8, Y=1/12 and Z=1/30. The supplied source's answer choices are corrupted in the extracted PDF (they appear as expressions such as 4^(4/29)), and the stated constraints do not specify whether the same pair must work for a full day or whether partial-day work is allowed. Because that detail changes the optimization, the answer key's 'second option' cannot be responsibly converted into a verified numeric answer from the supplied text alone.

Aman, Baman and Chaman alone finish in 15, 20 and 25 days. When working with somebody, their efficiencies reduce by 30%, 20% and 50%, respectively. Nobody may work three consecutive days. Find the maximum percentage completed in four days.

Worked solution
The individual rates are 1/15, 1/20 and 1/25. The wording does not fully define whether a worker's reduced efficiency applies whenever anyone else works that day, only to people on the same team, or how many people may work per day. These interpretations produce different schedules and totals. The source answer key states 34%, but the problem statement as supplied is not precise enough to independently establish that value.

A, B and C alone finish in 20, 30 and 60 days. A works every day and is assisted by B and C on every third day. Find the total time.

Worked solution
On ordinary days A completes 1/20. On every third day, the rate is 1/20+1/30+1/60=1/10. Each 3-day block completes 1/20+1/20+1/10=1/5. Five such blocks complete the job, requiring 15 days.

P+Q finish in 10 days and Q+R in 12 days. All three are paid ■300 in proportion to work; Q receives ■140. Find P's share.

Worked solution
Let rates be p,q,r. p+q=1/10 and q+r=1/12, so total rate p+q+r=11/60−q. Q's payment fraction is 140/300=7/15, so q/(11/60−q)=7/15. Solving gives q=7/120. Then p=1/10−7/120=1/24. Total rate = 1/8, so P's fraction is (1/24)/(1/8)=1/3. P receives ■300/3=■100.

Ten men start a job that would take them 15 days. After every three days, two more men join. On which day is the job completed?

Worked solution
Total work = 10×15=150 man-days. Days 1–3: 10×3=30, cumulative 30. Days 4–6: 12×3=36, cumulative 66. Days 7–9: 14×3=42, cumulative 108. Days 10–12: 16×3=48, cumulative 156. The job therefore finishes during day 12.

P and Q alternate days. With P starting, the job is finished in 17 days. With Q starting, it is finished in 53/3 days. Find Q's solo time.

Worked solution
With P starting for 17 days, P works 9 days and Q works 8: 9p+8q=1. With Q starting, the job finishes after 17 and 2/3 days: Q works 9 full days and P works 8 and 2/3 days. So 9q+(26/3)p=1. Multiply the second equation by 3: 27q+26p=3. The first multiplied by 3 is 27p+24q=3. Subtracting gives p=3q. Substitute into 9p+8q=1: 27q+8q=35q=1, so q=1/35.

Eight workers make eight motorcycles in eight days. How many motorcycles do four workers make in four days, assuming equal efficiency?

Worked solution
Eight workers for eight days supply 8×8=64 worker-days and make 8 motorcycles, so each motorcycle requires 8 worker-days. Four workers for four days supply 4×4=16 worker-days. They can make 16/8=2 motorcycles.